RAJALAKSHMI ENGINEERING COLLEGE

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M2

Started on	Monday, 28 September 2026, 10:08 PM
State	Finished
Completed on	Monday, 28 September 2026, 10:08 PM
Time taken	45 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int a[n];
    int dp[n];

    for(int i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
        dp[i] = 1;
    }

    int ans = 1;

    for(int i = 0; i < n; i++)
    {
        for(int j = 0; j < i; j++)
        {
            if(a[j] <= a[i])
            {
                if(dp[j] + 1 > dp[i])
                {
                    dp[i] = dp[j] + 1;
                }
            }
        }

        if(dp[i] > ans)
        {
            ans = dp[i];
        }
    }

    printf("%d", ans);

    return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔

	Input	Expected	Got	
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course